

Liquidity Risk Case

Overview

The Liquidity Risk Case challenges participants to evaluate the liquidity risk associated with tender offers across three securities. Participants will receive multiple tender offers throughout the case and must make appropriate judgements on profitability, acceptance, and execution. Profits are generated by exploiting price differentials between market prices and tendered prices. Once a tender is accepted, participants should efficiently close out their positions to maximize returns while minimizing liquidity and market risk.

Description

The case runs for 7 minutes (420 seconds). During that time, tender offers are generated and distributed at random intervals. Different participants may receive different offers at different times, but each participant receives an equal number of tenders with variations in price and quantity. No trading commission applies to tender offers. Data retrieval via Real-Time Data (RTD) Links and order submission using the RIT API are both enabled.

Types of Tender Offers

Each heat involves three types of tender offers:

- **Private Tenders:** Assigned directly to individual participants, offering the chance to buy or sell a fixed volume of shares in CRZY, TAME, or CROC at a price influenced by the current market price. Once accepted, the order is executed immediately from the participant's inventory at the specified price.
- **Competitive Auctions:** Distributed simultaneously to all traders, who must submit competitive yet profitable prices for a specified volume. Bids that exceed a hidden baseline reserve price are automatically filled, regardless of competing bids. If accepted, the transaction executes at the price submitted by the participant.
- **Winner-Take-All Tenders:** All participants submit bids or offers for a fixed volume. The tender is awarded to the highest bid (or lowest offer) that meets the hidden baseline reserve price. If no submission meets the reserve, the tender is void — for example, if all participants bid \$8.00 for a security with a \$10.00 reserve price, no trade is executed.

Market Dynamics

Participants will be able to trade 3 securities, namely CRZY, TAME, and CROC. The starting price, volatility, brief liquidity profile, and max trade size of each security is given below:

Security	CRZY	TAME	CROC
Start Price	\$ 10.00	\$ 25.00	\$ 20.00
Commissions	\$ 0.02	\$ 0.02	\$ 0.02
Volatility	High	Medium	High
Liquidity	Medium-low	Medium-high	Varied
Max Trade Size	25,000 shares	10,000 shares	20,000 shares

Prices follow a random-walk process, with returns drawn from a normal distribution centered near zero. This means the probability of prices moving up or down is equal and future prices cannot be reliably predicted — any directional trade is effectively a speculative bet.

Trading Limits and Transaction Costs

Each participant is subject to gross and net trading limits across all combined positions:

Type	Gross Limit	Net Limit
LIMIT-STOCK	250,000 shares	100,000 shares

Trading limits are enforced; participants cannot exceed them.

Speculation and Front-Running Penalties

This case is designed exclusively to reward participants for correctly identifying, accepting, and efficiently closing out tender-related positions while managing liquidity and execution risk. All other strategies are discouraged.

A **speculative trade** is any trade that opens a *new* position unrelated to closing out an accepted tender. Since prices follow a random walk — equally likely to rise or fall — any such trade amounts to a directional bet with no informational edge will be penalized.

Similarly, each tender comes with a short decision window that may vary by security. For example, a participant might receive a tender at $t = 0$ and have until $t = 30$ to accept or reject it. Any trades executed during this window, before the tender decision is finalized, are flagged as **front-running** — the participant is acting on advance knowledge of a pending order they have not yet committed to. Front-running is penalized identically to speculative trading.

To be clear: only trades that **reduce** an existing position from an accepted tender are permitted without penalty. Opening any new position — whether in anticipation of a tender, in parallel with one, or independently — will incur the following penalties:

- **First 5,000 shares of new position:** \$0.20 per share
- **Beyond 5,000 shares:** \$0.40 per share

Adjusted P&L including total fines paid is displayed in the **Trader Info** section of the RIT application. Speculation fines can also be reviewed in the **Transaction Log**.

Position Close-Out

All open positions are closed at the end of the trading case based on the last traded price. Computerized market makers increase liquidity near the end of trading to ensure closing prices cannot be easily manipulated.

Key Objectives

- **Objective 1:** Evaluate the profitability of tender offers by analyzing market liquidity from the limit order book. Accept tenders expected to generate positive profits; reject unattractive offers.
- **Objective 2:** Use a combination of limit orders, market orders, and marketable limit orders to mitigate liquidity and price risk from holding open positions. Markets can move against your transaction prices, so large open positions carry meaningful downside risk.